

```
import pandas as pd
import numpy as np

df
=pd.read_csv("C:/Users/admin/Downloads/archive/gender_submission.csv")

df.head
```

		PassengerId	Survived
0	892	0	
1	893	1	
2	894	0	
3	895	0	
4	896	1	
...	...	...	
413	1305	0	
414	1306	1	
415	1307	0	
416	1308	0	
417	1309	0	

```
[418 rows x 2 columns]>
```

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv")

df.head
```

	5.1	3.5	1.4	0.2	Iris-setosa
0	4.9	3.0	1.4	0.2	Iris-setosa
1	4.7	3.2	1.3	0.2	Iris-setosa
2	4.6	3.1	1.5	0.2	Iris-setosa
3	5.0	3.6	1.4	0.2	Iris-setosa
4	5.4	3.9	1.7	0.4	Iris-setosa
...	...	...	...	...	...
144	6.7	3.0	5.2	2.3	Iris-virginica
145	6.3	2.5	5.0	1.9	Iris-virginica
146	6.5	3.0	5.2	2.0	Iris-virginica
147	6.2	3.4	5.4	2.3	Iris-virginica
148	5.9	3.0	5.1	1.8	Iris-virginica

```
[149 rows x 5 columns]>
```

```
df.tail
```

	5.1	3.5	1.4	0.2	Iris-setosa
0	4.9	3.0	1.4	0.2	Iris-setosa
1	4.7	3.2	1.3	0.2	Iris-setosa
2	4.6	3.1	1.5	0.2	Iris-setosa
3	5.0	3.6	1.4	0.2	Iris-setosa
4	5.4	3.9	1.7	0.4	Iris-setosa

```

..      ...      ...      ...      ...      ...
144  6.7  3.0  5.2  2.3  Iris-virginica
145  6.3  2.5  5.0  1.9  Iris-virginica
146  6.5  3.0  5.2  2.0  Iris-virginica
147  6.2  3.4  5.4  2.3  Iris-virginica
148  5.9  3.0  5.1  1.8  Iris-virginica

```

```
[149 rows x 5 columns]>
```

```
df.isnull()
```

```

      5.1      3.5      1.4      0.2  Iris-setosa
0  False  False  False  False  False
1  False  False  False  False  False
2  False  False  False  False  False
3  False  False  False  False  False
4  False  False  False  False  False
..      ...      ...      ...      ...      ...
144 False  False  False  False  False
145 False  False  False  False  False
146 False  False  False  False  False
147 False  False  False  False  False
148 False  False  False  False  False

```

```
[149 rows x 5 columns]
```

```
df.isnull().sum
```

```
<bound method NDFrame._add_numeric_operations.<locals>.sum of
```

```

5.1      3.5      1.4      0.2  Iris-setosa
0  False  False  False  False  False
1  False  False  False  False  False
2  False  False  False  False  False
3  False  False  False  False  False
4  False  False  False  False  False
..      ...      ...      ...      ...      ...
144 False  False  False  False  False
145 False  False  False  False  False
146 False  False  False  False  False
147 False  False  False  False  False
148 False  False  False  False  False

```

```
[149 rows x 5 columns]>
```

```
df.describe
```

```
<bound method NDFrame.describe of      5.1  3.5  1.4  0.2      Iris-
setosa
```

```

0      4.9  3.0  1.4  0.2      Iris-setosa
1      4.7  3.2  1.3  0.2      Iris-setosa
2      4.6  3.1  1.5  0.2      Iris-setosa

```

```

3      5.0  3.6  1.4  0.2      Iris-setosa
4      5.4  3.9  1.7  0.4      Iris-setosa
..      ...  ...  ...  ...      ...
144    6.7  3.0  5.2  2.3      Iris-virginica
145    6.3  2.5  5.0  1.9      Iris-virginica
146    6.5  3.0  5.2  2.0      Iris-virginica
147    6.2  3.4  5.4  2.3      Iris-virginica
148    5.9  3.0  5.1  1.8      Iris-virginica

```

```
[149 rows x 5 columns]>
```

```
df.columns
```

```
Index(['5.1', '3.5', '1.4', '0.2', 'Iris-setosa'], dtype='object')
```

```
df.dtypes
```

```

5.1      float64
3.5      float64
1.4      float64
0.2      float64
Iris-setosa  object
dtype: object

```

```
df.shape
```

```
(149, 5)
```

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv",
header = None)
```

```
col_names =
```

```
['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width', 'Species']
```

```
iris = pd.read_csv('https://archive.ics.uci.edu/ml/machine-learning-
databases/iris/iris.data', names = col_names)
```

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv",
header = None)
```

```
df.head
```

```

<bound method NDFrame.head of      0      1      2      3      4
0      5.1  3.5  1.4  0.2      Iris-setosa
1      4.9  3.0  1.4  0.2      Iris-setosa
2      4.7  3.2  1.3  0.2      Iris-setosa
3      4.6  3.1  1.5  0.2      Iris-setosa
4      5.0  3.6  1.4  0.2      Iris-setosa
..      ...  ...  ...  ...      ...
145    6.7  3.0  5.2  2.3      Iris-virginica
146    6.3  2.5  5.0  1.9      Iris-virginica
147    6.5  3.0  5.2  2.0      Iris-virginica

```

```
148  6.2  3.4  5.4  2.3  Iris-virginica
149  5.9  3.0  5.1  1.8  Iris-virginica
```

```
[150 rows x 5 columns]>
```

```
col_names =
['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width', 'Species']
```

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv",
names = col_names)
```

```
df.head
```

```
<bound method NDFrame.head of
Petal_Length  Petal_Width  of      Sepal_Length  Sepal_Width
0              5.1          3.5      1.4          0.2      Iris-
setosa
1              4.9          3.0      1.4          0.2      Iris-
setosa
2              4.7          3.2      1.3          0.2      Iris-
setosa
3              4.6          3.1      1.5          0.2      Iris-
setosa
4              5.0          3.6      1.4          0.2      Iris-
setosa
..              ...          ...      ...          ...
...
145            6.7          3.0      5.2          2.3      Iris-
virginica
146            6.3          2.5      5.0          1.9      Iris-
virginica
147            6.5          3.0      5.2          2.0      Iris-
virginica
148            6.2          3.4      5.4          2.3      Iris-
virginica
149            5.9          3.0      5.1          1.8      Iris-
virginica
```

```
[150 rows x 5 columns]>
```

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv",
names = col_names)
```

```
df.isnull()
```

```
      Sepal_Length  Sepal_Width  Petal_Length  Petal_Width  Species
0              False          False          False          False  False
1              False          False          False          False  False
2              False          False          False          False  False
3              False          False          False          False  False
4              False          False          False          False  False
```

145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

[150 rows x 5 columns]

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv",
names = col_names)
```

```
df.isnull()
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

[150 rows x 5 columns]

```
df =pd.read_csv("C:/Users/admin/Downloads/archive (1)/iris.data.csv",
names = col_names)
```

```
df.isnull()
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	True	True	True	True

[150 rows x 5 columns]

```
df.columns
```

```
Index(['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width',  
      'Species'],  
      dtype='object')
```

```
df.isnull().sum()
```

```
Sepal_Length    0  
Sepal_Width     1  
Petal_Length    5  
Petal_Width     1  
Species         1  
dtype: int64
```

```
from sklearn import preprocessing
```

```
df['Species'].unique()
```

```
array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica', nan],  
      dtype=object)
```

```
label_encoder = preprocessing.LabelEncoder()
```

```
df['Species']= label_encoder.fit_transform(df['Species'])
```

```
df['Species'].unique()
```

```
array([0, 1, 2, 3])
```

```
df.dtypes
```

```
Sepal_Length    float64  
Sepal_Width     float64  
Petal_Length    float64  
Petal_Width     float64  
Species         int32  
dtype: object
```

```
df.head()
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
df.tail()
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2

148	6.2	3.4	5.4	2.3	2
149	5.9	NaN	NaN	NaN	3